8 **(a)** A beam of electrons is directed along a normal towards a barrier, as shown in Fig. 8.1.

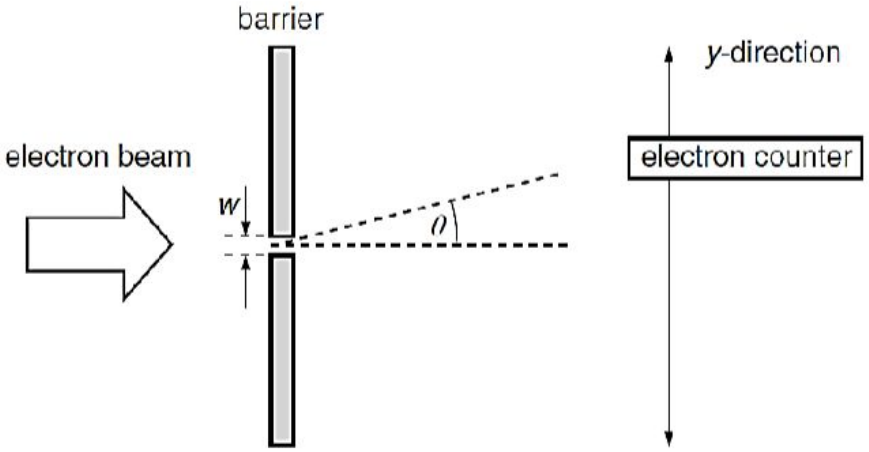


Fig. 8.1

The barrier contains a single slit of width w. Beyond the slit there is a counter that counts electrons. This can be moved in the y-direction to compare the rate of arrival of electrons at different values of the angle θ from the original direction of the beam.

(i) Louis de Broglie suggested that particles such as electrons have an associated wavelength. State the equation for the de Broglie wavelength of an electron and identify each term.

.....

$\lambda = h/p$,

where λ is the de Broglie wavelength, h is the Planck constant, and p is the momentum of the electron. [1]

.....

..... [1]

(ii) State the condition for diffraction to be observable in this scenario.

The de Broglie wavelength of the electrons is comparable to the slit width w. [1]

.....

..... [1]

(iii) Electrons are accelerated by an accelerating voltage. State and explain what happens to angle θ when magnitude of the accelerating voltage decreases.

When the accelerating voltage is decreased, the electrons (loses less electric potential energy and) gains less kinetic energy; hence, its speed v is decreased and momentum, (the product of its mass m and v) decreases. Since de Broglie's wavelength is inversely proportional to its momentum, wavelength increases. [1]

The electrons will be more diffracted ($d\sin\theta = n\lambda$) and hence the angle θ increases. [1]

..... [2]

(b) Werner Heisenberg used a different approach involving what is now known as the uncertainty principle. This can also be used to explain why the electron beam spreads out after passing through the slit. One version of this involves the equation

$$\Delta y \Delta p \gtrsim h \, .$$

Using the uncertainty principle, explain what happens to angle θ when the slit is made narrower.

A narrower slit means a smaller value of w, so Δy is smaller. [1]

By the uncertainty principle $\Delta y \Delta p \gtrsim h$, a smaller Δy would mean a larger Δp, meaning the uncertainty in the (y-component of) momentum is greater. [1]

A larger spread in y-momentum means electrons travel across a wider angles θ, so the beam is spread out more. [1]

..... [3]

(c) Explain whether your answer in **(b)** is consistent with single slit diffraction, when the slit is made narrower.

.....

In single slit diffraction, sin $\theta = \lambda / b$, the angle θ is inversely proportional to slit width, when wavelength is constant. So, it results in a larger angle θ. [1]

By the uncertainty principle, a narrower slit implies a larger spread in y-momentum means electrons travel across a wider range of angles, so the beam is spread out more it results in a larger angle θ. [1]

The answer is consistent with single slit diffraction.

.....

.....

.....

.....

.....

..... [2]

(d) Fig. 8.2 shows a cooler region of hydrogen gas surrounding a hot gas cloud emitting white light.

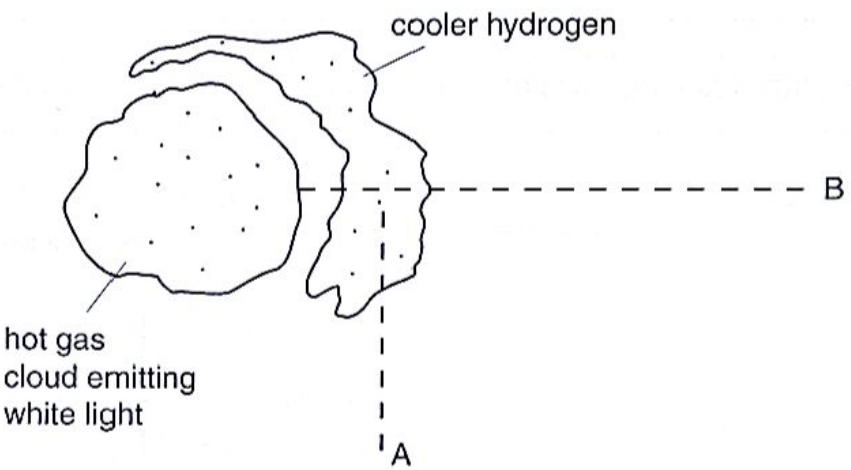


Fig. 8.2

		(i)	Describe the difference in appearance between an emission and absorption line spectra.
			..... Emission line spectra consist of discrete bright lines of different colours on a dark background ... Absorption line spectra are continuous spectrum crossed by dark lines. ... [1] .... [1]
		(ii)	State the types of spectra seen from point A and B.
			1. point A: ..... A: Emission line spectra ..... **2.** point B: ..... B: Absorption spectra ..... [1]
		(iii)	Explain how spectra seen at point B is obtained. Energy levels in gas atoms are discrete. [1] ... When white light passes through the cooler hydrogen, only photons that has energy equal to the difference between two energy levels in the gas atom will be absorbed to cause electron excitation. [1] The unabsorbed photons will form a continuous spectrum of light on the screen with discrete dark lines. ... Electrons in the excited state are unstable and will de-excite to ground state to produce photons with energy equal to the difference between the excited state and lower state (accept ground state). These photons are emitted in random direction and will be missing from the spectra at point B. [1] ... . [3]

(e) The energy levels E_n in a hydrogen atom are given by the empirical equation

$$E_n = \frac{-13.6 \text{ eV}}{n^2},$$

where n is an integer.

The four visible spectral lines P, Q, R and S in the hydrogen spectrum are illustrated in Fig. 8.3.

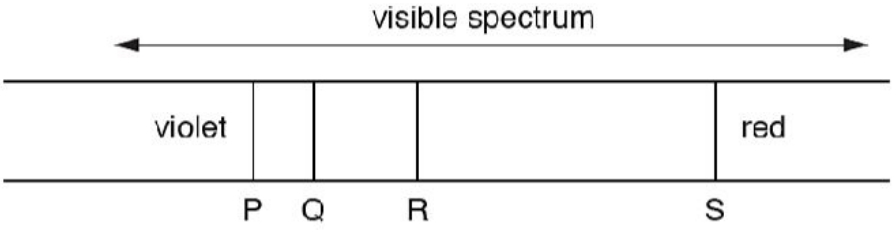


Fig. 8.3

The frequencies of three of these spectral lines are shown in the table.

spectral line	frequency / 10^{12} Hz
P	731
Q	f_Q
R	617
S	456

These frequencies are emitted when the energy level to which the electrons arrive is for $n = 2$.

(i) Deduce the value of n for the energy level from which an electron has moved in order to produce spectral line S.

$$E_n - E_2 = hf$$

[1]

$$\left(\frac{-13.6 \text{ eV}}{n^2}\right) - \left(\frac{-13.6 \text{ eV}}{2^2}\right) = \left(6.63 \times 10^{-34}\right)\left(456 \times 10^{12}\right)$$

$$\left(-13.6 \times 1.60 \times 10^{-19} \text{ J}\right)\left(\frac{1}{n^2} - \frac{1}{4}\right) = 3.0233 \times 10^{-19}$$

[1]

$$\left(\frac{1}{n^2} - \frac{1}{4}\right) = -0.139$$

$$n = 3$$

[1]

n = ... [3]

(ii) Determine the frequency of the spectral line Q, f_Q.

$$E_5 - E_2 = hf_Q \qquad [1]$$

$$\left(\frac{-13.6\ \text{eV}}{5^2}\right) - \left(\frac{-13.6\ \text{eV}}{2^2}\right) = \left(6.63 \times 10^{-34}\right) f_Q \qquad [1]$$

$$\left(-13.6 \times 1.60 \times 10^{-19}\ \text{J}\right)\left(\frac{1}{25} - \frac{1}{4}\right) = \left(6.63 \times 10^{-34}\right) f_Q$$

$$f_Q = 689 \times 10^{12}\ \text{Hz} \qquad [1]$$

f_Q = ... Hz [3]

[Total: 20]